

B.C.A. (Semester-II) Examination, 2020

(Session : 2019-22)

COMPUTER ARCHITECTURE

[Paper Code : BC-202]

Time : Three Hours

[Maximum Marks : 80]

Note: Candidates are required to give their answers in their own words as far as practicable. The questions are of the equal value. Answer any five questions.

1. Convert the following in Binary Number and perform the arithmetic operation using 1's/2's complement. Indicate overflow/underflow (if any).

(a) $(6)_{10} + (-13)_{10}$

(b) $(45)_{10} + (-15)_{10}$

2. Using 10's complement, find:

(a) $72532 - 3250$

(b) $3250 - 72532$

3. What do you mean by Logic Gates? Briefly explain each of them with respective diagram and Truth Table.

4. State Demorgan's Theorem and provide basic properties of Boolean Algebra.

5. What do you mean by Digital Counters? Design a 3 bit binary counter using Flip-Flop.

6. What is Flip-Flop? Name them. Explain R.S Flip-Flop with diagram and Truth Table.

7. What do you mean by Data Transfer? Differentiate Synchronous and Asynchronous mode of data transfer.

8. Explain I/O Interface. Briefly explain DMA and I/O Processor.

9. What do you mean by Memory? Classify and introduce them. Differentiate RAM and ROM with examples.

10. Write short notes on any two of the following:

(a) Sequential and Combinational Circuit

(b) Flip-Flop and Latches

(c) Register and Counter

(d) Memory Organization